

CariSurg MedTech Pathways 2026

Week 7 Cost-Benefit Memo

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GitHub Repository

<https://github.com/baldiea75-tech/carisurg-portfolio.git>

Verdict

Stay with the Week 6 logistic regression model for Phase 3. A random forest, tested at default settings and through a wider search across tree depth and class weighting, cannot be found to beat logistic regression on both accuracy and recall for the most critical patients at the same time. Every gain on the rare, most urgent class was paid for somewhere else, in accuracy, in precision or in both.

Methodology

This analysis reuses the exact Week 6 setup so the comparison is fair. The target is ESI level, the same 208 features (vital signs and presenting complaint flags, with leakage, demographic and administrative columns excluded) and the same 80 percent training, 20 percent test split with random seed 42 were used throughout, confirmed against the Week 6 test set counts exactly.

A random forest, an ensemble of decision trees that vote together on each prediction, was chosen as the added complexity for Phase 3, ahead of gradient boosting or a small neural network, because it extends naturally from the Week 6 decision tree and still supports a feature importance explanation. A first random forest was trained at default-style settings. A second version added class weighting to test whether the rare class problem could be fixed directly. A wider search was then run across tree depth and several weighting schemes to check whether a better trade-off existed than the first two attempts found.

Overall Benchmark Table

Model	Accuracy	Macro F1	Weighted F1	Recall (ESI 1)
Dummy (baseline)	0.375	0.204	0.375	0.000
Logistic Regression	0.667	0.492	0.661	0.250
Decision Tree	0.556	0.216	0.463	0.000
Random Forest	0.580	0.238	0.500	0.000
Random Forest (balanced)	0.478	0.369	0.481	0.250
Random Forest (best trade-off found)	0.470	0.367	0.476	0.562

Sixteen patients in the test set were truly ESI 1. Logistic regression correctly flagged 4 of them. The best random forest configuration found, maximum depth of 8 with balanced class weighting, flagged 9 of them, more than doubling logistic regression's recall. That gain came from flagging 102 patients as ESI 1 in total, so only 9 of every 102 flags were correct, a precision of 0.088, and overall accuracy fell to 0.470 from logistic regression's 0.667.

Model Evaluation Graphs

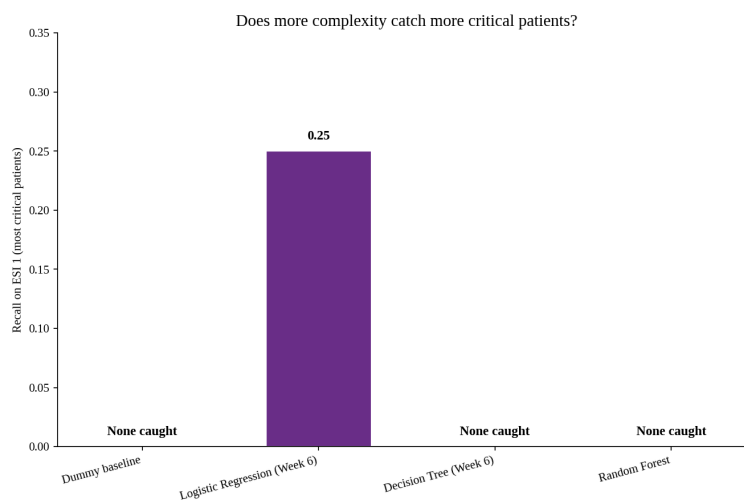


Figure 1. chart compares recall on ESI 1 across the Week 6 baselines and the default-settings random forest. The unweighted random forest catches none of the sixteen true ESI 1 patients in the test set, matching the decision tree and doing worse than logistic regression's one in four.

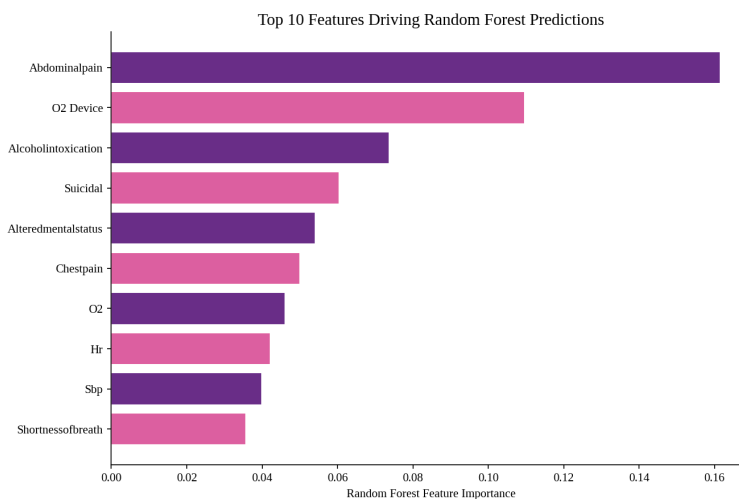


Figure 2. Chart shows feature importance describes the random forest as a whole rather than any single prediction. Abdominal pain, use of an oxygen delivery device and alcohol intoxication are the three strongest drivers overall, each clinically plausible as a signal the model is using.

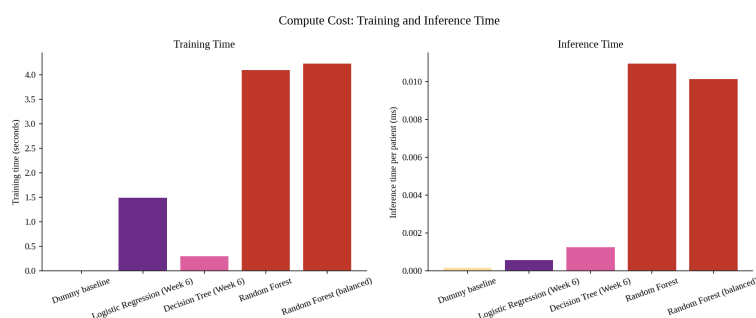


Figure 3. shows training time and inference time were both measured with Python's `time.perf_counter`, timed identically for every model in the same run. The random forest costs roughly three times as long to train and around seventeen times as long to run per prediction compared with logistic regression.

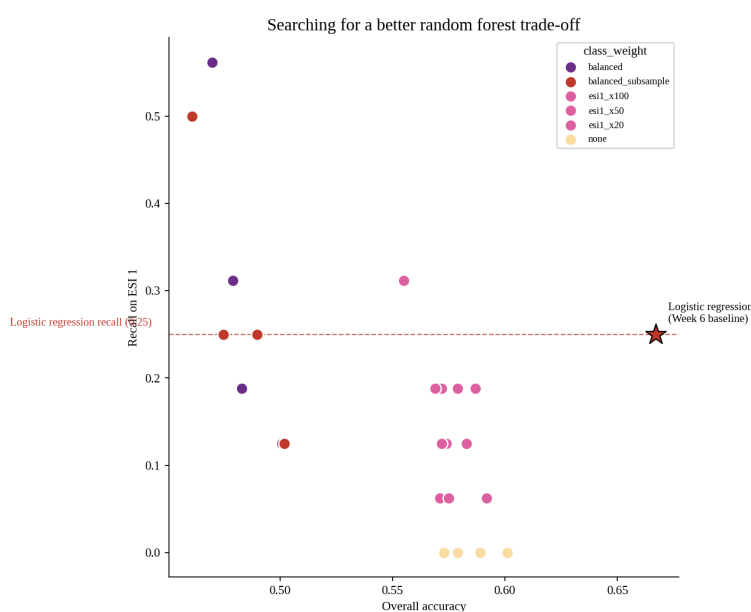


Figure 4. Shows every configuration tested across tree depth and class weighting, plotted as accuracy against recall on ESI 1. The Week 6 logistic regression baseline is marked with a star. No point sits above and to the right of that star at once, confirming that no configuration found beats logistic regression on both accuracy and recall together.

Three Arguments For the Random Forest

First, it uses more of the available signal than a linear model can, since it is the only model here capable of capturing non-linear combinations of symptoms rather than a single straight line boundary between classes.

Second, class weighting gives it a lever a plain decision tree does not have. With the right settings it can be pushed to catch more than twice as many ESI 1 patients as logistic regression.

Third, feature importance still provides a defensible, model-level audit trail. Abdominal pain, oxygen device use and altered mental status are all clinically sensible top drivers, even without a single-patient explanation.

Three Arguments Against the Random Forest

First, no configuration tested beats logistic regression on both accuracy and recall at once. The best trade-off found catches more ESI 1 patients only by giving up nearly twenty points of overall accuracy and by flagging so many false alarms that precision on that class falls to 0.088.

Second, it costs substantially more to run. Roughly three times the training time and seventeen times the inference time per patient compared with logistic regression, for a model that does not clearly outperform it.

Third, it cannot be explained to a clinician for a single patient in under a minute. A random forest prediction is the combined vote of many trees, not a short list of yes or no questions the way a decision tree or a logistic regression score can be.

Risks and Unknowns

The search covered a reasonable range of tree depths and class weighting schemes but was not exhaustive; a different combination might yet yield a better point on the trade-off curve. Gradient boosting and a small neural network remain untested and might behave differently on the rare class problem. All training and inference times were measured on a single machine in a single run and would need to be remeasured on Mercer General's actual infrastructure before being used in a real capacity planning conversation.

Recommendation

Recommend Phase 3 continues with the Week 6 logistic regression model rather than adopting a random forest. The wider search this week shows the rare class problem can be pushed on directly, catching more than twice as many ESI 1 patients is possible, but only at a cost in accuracy and precision that has not yet been shown to be worthwhile. The more promising next step is applying that same class weighting idea to logistic regression itself, rather than changing model family, since logistic regression already outperforms every random forest configuration tested on overall accuracy while remaining the easiest model of the three to explain to a clinician.